Referential Integrity

Theory & Practical

Theory

What are Cascading Referential Integrity Constraints in SQL Server?

1. **Cascading Referential Integrity Constraints** in **SQL Server** are rules applied on **FOREIGN KEY** relationships that automatically perform actions on child (referencing) rows when a parent (referenced) row is **updated** or **deleted**.

2. These help maintain **data consistency** without writing manual triggers or code.

What Are Cascading Actions?

When creating a **FOREIGN KEY**, you can specify what SQL Server should do if the parent row is:

Deleted → ON DELETE
Updated → ON UPDATE

In Short:
ON DELETE/UPDATE Action_Name

SQL Server provides 4 cascading options:

1. NO ACTION
2. SET NULL
3. SET DEFAULT
4. CASCADE

NO ACTION

- Default behaviour

SET NULL

- When the parent row is deleted or updated,

The foreign key column in child table is set to NULL

Important Condition

The foreign key column must allow NULL

SET DEFAULT

- Child table foreign key is set to its DEFAULT value when the parent changes

Important Condition

Column must have DEFAULT constraint

Default value must exist in parent table

CASCADE

- Changes in parent table automatically apply to child table

Behavior

DELETE → child rows deleted automatically

UPDATE → foreign key updated automatically

Identity Column

- **Note:**

Identity Column value not update

see below example

```sql
CREATE TABLE Test(
    id INT IDENTITY(1,1) PRIMARY KEY,
    fname VARCHAR(10)
);

INSERT INTO Test (fname) VALUES ('a'),('b'),('c'),('d');

select * from Test;
```

Results | Messages

	id	fname
1	1	a
2	2	b
3	3	c
4	4	d

```sql
23  update Test
24  set id = 100
25  where id = 1;
26
```

No issues found

Messages

```
Msg 8102, Level 16, State 1, Line 23
Cannot update identity column 'id'.

Completion time: 2025-12-06T21:55:39.1382304+05:30
```

Practical

```sql
--Step:1
--create data base
CREATE DATABASE [Referential Integrity];

--use data base
USE [Referential Integrity];

SET ANSI_NULLS ON
GO

SET QUOTED_IDENTIFIER ON
GO
```

NO ACTION

```sql
--Set:2

CREATE TABLE Parent(
    Parent_Id INT IDENTITY(1,1),
    Parent_Code NVARCHAR(5)
        CONSTRAINT PK_Tb_Parent_Col_Parent_Code PRIMARY KEY,
    Parent_Name NVARCHAR(15),
    Parent_Email NVARCHAR(20)
);
```

```sql
CREATE TABLE Child(
    Child_Id INT IDENTITY(1,1),
    Child_Name NVARCHAR(15),
    Child_Email NVARCHAR(20),
    Parent_Code NVARCHAR(5)
        CONSTRAINT FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code
        FOREIGN KEY (Parent_Code) REFERENCES Parent(Parent_Code)
);
```

Results | Messages

PK

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	1	P-101	P1	p1@gmail.com
2	Parent	2	P-102	P2	p2@gmail.com
3	Parent	3	P-103	P3	p3@gmail.com
4	Parent	4	P-104	P4	p4@gmail.com

FK

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	P-101
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	P-101
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	P-101
7	Child	7	C7	c7@gmail.com	P-104

```sql
59 DELETE FROM Parent WHERE Parent_id = 1;
```

% 12 0 Ln: 59 Ch: 40 TABS

Messages

```
Msg 547, Level 16, State 0, Line 59
The DELETE statement conflicted with the REFERENCE constraint
"FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code".
The conflict occurred in database "Referential Integrity",
table "dbo.Child", column 'Parent_Code'.
```

```sql
65 UPDATE Parent
66 SET Parent_Code = 'P-501'
67 WHERE Parent_id = 1;
```

% 12 0

Messages

```
Msg 547, Level 16, State 0, Line 65
The UPDATE statement conflicted with the REFERENCE constraint
"FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code".
 The conflict occurred in database "Referential Integrity",
 table "dbo.Child", column 'Parent_Code'.
```

```sql
CREATE TABLE Child(
    Child_Id INT IDENTITY(1,1),
    Child_Name NVARCHAR(15),
    Child_Email NVARCHAR(20),
    Parent_Code NVARCHAR(5)
        CONSTRAINT FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code
        FOREIGN KEY (Parent_Code) REFERENCES Parent(Parent_Code)
);
```

Table -1

```sql
CREATE TABLE Child(
    Child_Id INT IDENTITY(1,1),
    Child_Name NVARCHAR(15),
    Child_Email NVARCHAR(20),
    Parent_Code NVARCHAR(5)
        CONSTRAINT FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code
        FOREIGN KEY (Parent_Code) REFERENCES Parent(Parent_Code)
        ON DELETE NO ACTION
        ON UPDATE NO ACTION
);
```

Table -2

Both Table Same because No Action is default.

Above error

So Parent table row not deleted or

not updated because mapping

with child table

1st step Child table row delete

and

2nd step Parent table row delete.

SET NULL

Case:1 FK Column NOT NULL

```sql
--Step: 3 SET NULL use but FK COLUMN NOT NULL
--Parent table same only child table changes
--so child table delete and create few modification but same data
```

```sql
DROP TABLE Child;

CREATE TABLE Child(
    Child_Id INT IDENTITY(1,1),
    Child_Name NVARCHAR(15),
    Child_Email NVARCHAR(20),
    Parent_Code NVARCHAR(5) NOT NULL
        CONSTRAINT FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code
        FOREIGN KEY (Parent_Code) REFERENCES Parent(Parent_Code)
        ON DELETE SET NULL
        ON UPDATE SET NULL
);
```

column Not NULL and cascading SET NULL

not posible

Messages

```
Msg 1761, Level 16, State 0, Line 110
Cannot create the foreign key "FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code" with
the SET NULL referential action, because one or more referencing columns are not nullable.
Msg 1750, Level 16, State 1, Line 110
Could not create constraint or index. See previous errors.
```

SET NULL

Case:2 FK Column NULL

```sql
--Step: 4 SET NULL use but now FK COLUMN NULL
--Parent table same only child table changes
--so child table delete and create few modification but same data

DROP TABLE Child;

CREATE TABLE Child(
    Child_Id INT IDENTITY(1,1),
    Child_Name NVARCHAR(15),
    Child_Email NVARCHAR(20),
    Parent_Code NVARCHAR(5) NULL
        CONSTRAINT FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code
        FOREIGN KEY (Parent_Code) REFERENCES Parent(Parent_Code)
        ON DELETE SET NULL
        ON UPDATE SET NULL
);
```

```sql
SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	1	P-101	P1	p1@gmail.com
2	Parent	2	P-102	P2	p2@gmail.com
3	Parent	3	P-103	P3	p3@gmail.com
4	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	P-101
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	P-101
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	P-101
7	Child	7	C7	c7@gmail.com	P-104

```sql
DELETE  FROM Parent WHERE Parent_Id = 1;

SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	P-102	P2	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	NULL
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	NULL
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	NULL
7	Child	7	C7	c7@gmail.com	P-104

this row delete

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	1	P-101	P1	p1@gmail.com
2	Parent	2	P-102	P2	p2@gmail.com
3	Parent	3	P-103	P3	p3@gmail.com
4	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	P-101
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	P-101
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	P-101
7	Child	7	C7	c7@gmail.com	P-104

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	P-102	P2	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	NULL
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	NULL
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	NULL
7	Child	7	C7	c7@gmail.com	P-104

```sql
UPDATE Parent
SET Parent_Name = 'p500'
where Parent_Id = 2

SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	P-102	p500	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	NULL
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	NULL
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	NULL
7	Child	7	C7	c7@gmail.com	P-104

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	P-102	P2	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	NULL
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	NULL
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	NULL
7	Child	7	C7	c7@gmail.com	P-104

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	P-102	p500	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	NULL
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	NULL
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	NULL
7	Child	7	C7	c7@gmail.com	P-104

Here Parent-Name

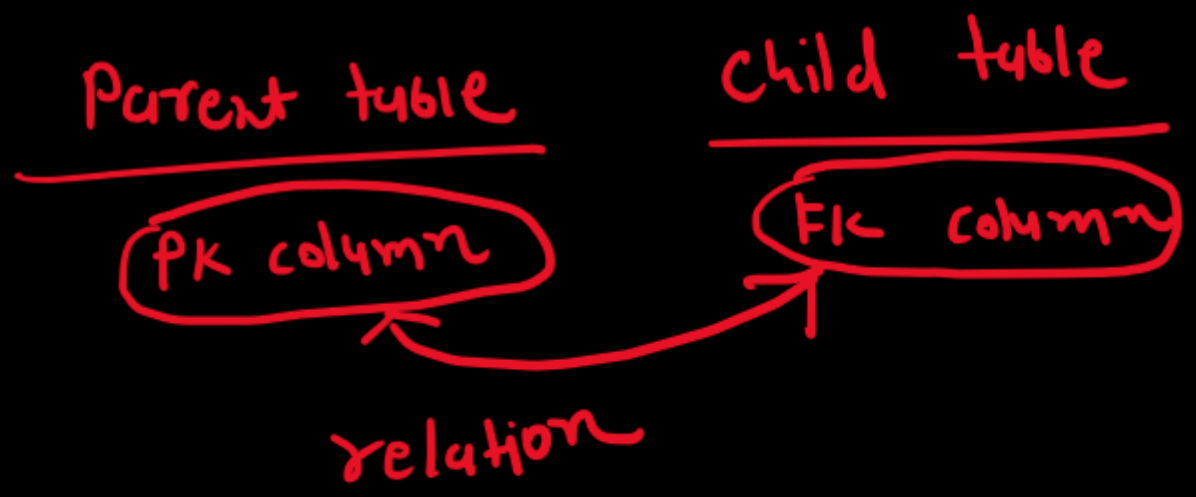
old → P2
new → P500

Only Parent table Parent-code value P-102 this row Parent-Name update.

FK table Parent_code

not any changes

because



only pk column value chang

↓

that's effect

↓

fk column.

```sql
UPDATE Parent
SET Parent_Code = 'P-501'
where Parent_Id = 2

SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	3	P-103	P3	p3@gmail.com
2	Parent	4	P-104	P4	p4@gmail.com
3	Parent	2	P-501	p500	p2@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	NULL
2	Child	2	C2	c2@gmail.com	NULL
3	Child	3	C3	c3@gmail.com	NULL
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	NULL
6	Child	6	C6	c6@gmail.com	NULL
7	Child	7	C7	c7@gmail.com	P-104

update P-501

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	P-102	p500	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	NULL
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	NULL
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	NULL
7	Child	7	C7	c7@gmail.com	P-104

NULL

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	3	P-103	P3	p3@gmail.com
2	Parent	4	P-104	P4	p4@gmail.com
3	Parent	2	P-501	p500	p2@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	NULL
2	Child	2	C2	c2@gmail.com	NULL
3	Child	3	C3	c3@gmail.com	NULL
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	NULL
6	Child	6	C6	c6@gmail.com	NULL
7	Child	7	C7	c7@gmail.com	P-104

SET DEFAULT

```sql
--Step: 5 SET DEFAULT
--now Parent table and child table both table drop and both table same data with created
--only child table modification

DROP TABLE Child;
DROP TABLE Parent;
```

```sql
CREATE TABLE Parent(
    Parent_Id INT IDENTITY(1,1),
    Parent_Code NVARCHAR(5)
        CONSTRAINT PK_Tb_Parent_Col_Parent_Code PRIMARY KEY,
    Parent_Name NVARCHAR(15),
    Parent Email NVARCHAR(20)
```

```sql
CREATE TABLE Child(
    Child_Id INT IDENTITY(1,1),
    Child_Name NVARCHAR(15),
    Child_Email NVARCHAR(20),

    Parent_Code NVARCHAR(5)
        CONSTRAINT DF_Tb_Child_Col_Parent_Code
            DEFAULT 'DF-01',

        CONSTRAINT FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code
        FOREIGN KEY (Parent_Code) REFERENCES Parent(Parent_Code)
        ON DELETE SET DEFAULT
        ON UPDATE SET DEFAULT
);
```

```sql
SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	1	P-101	P1	p1@gmail.com
2	Parent	2	P-102	P2	p2@gmail.com
3	Parent	3	P-103	P3	p3@gmail.com
4	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	P-101
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	P-101
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	P-101
7	Child	7	C7	c7@gmail.com	P-104

```sql
DELETE FROM Parent WHERE Parent_Id = 1;
```

```
Msg 547, Level 16, State 0, Line 229
The DELETE statement conflicted with the FOREIGN KEY constraint "FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code".
The conflict occurred in database "master", table "dbo.Parent", column 'Parent_Code'.
```

```sql
UPDATE Parent
SET Parent_Code = 'U-501'
WHERE Parent_Id = 1;
```

Messages

```
Msg 547, Level 16, State 0, Line 231
The UPDATE statement conflicted with the FOREIGN KEY constraint "FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code".
The conflict occurred in database "master", table "dbo.Parent", column 'Parent_Code'.
```

Above both delete & update error occur because child table set default value not existing parent table PK column.

```sql
--above delete / update error so 2 solution
--sol:1 child table only existing value set
--sol:2 insert default value in parent table

--i am going sol:2

INSERT INTO Parent (Parent_Name,Parent_Code,Parent_Email)
VALUES ('Default','DF-01','DF@gmail.com');

SELECT * FROM Parent ORDER BY Parent_Id;
```

Results | Messages

	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	1	P-101	P1	p1@gmail.com
2	2	P-102	P2	p2@gmail.com
3	3	P-103	P3	p3@gmail.com
4	4	P-104	P4	p4@gmail.com
5	5	DF-01	Default	DF@gmail.com

```sql
SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	5	DF-01	Default	DF@gmail.com
2	Parent	1	P-101	P1	p1@gmail.com
3	Parent	2	P-102	P2	p2@gmail.com
4	Parent	3	P-103	P3	p3@gmail.com
5	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	P-101
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	P-101
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	P-101
7	Child	7	C7	c7@gmail.com	P-104

```sql
DELETE FROM Parent WHERE Parent_Id = 1;

SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	5	DF-01	Default	DF@gmail.com
2	Parent	2	P-102	P2	p2@gmail.com
3	Parent	3	P-103	P3	p3@gmail.com
4	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	DF-01
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	DF-01
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	DF-01
7	Child	7	C7	c7@gmail.com	P-104

delete

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	5	DF-01	Default	DF@gmail.com
2	Parent	1	P-101	P1	p1@gmail.com
3	Parent	2	P-102	P2	p2@gmail.com
4	Parent	3	P-103	P3	p3@gmail.com
5	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	P-101
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	P-101
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	P-101
7	Child	7	C7	c7@gmail.com	P-104

Set default value

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	5	DF-01	Default	DF@gmail.com
2	Parent	2	P-102	P2	p2@gmail.com
3	Parent	3	P-103	P3	p3@gmail.com
4	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	DF-01
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	DF-01
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	DF-01
7	Child	7	C7	c7@gmail.com	P-104

```sql
UPDATE Parent
SET Parent_Code = 'U-501'
WHERE Parent_Id = 2;

SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	5	DF-01	Default	DF@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com
4	Parent	2	U-501	P2	p2@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	DF-01
2	Child	2	C2	c2@gmail.com	DF-01
3	Child	3	C3	c3@gmail.com	DF-01
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	DF-01
6	Child	6	C6	c6@gmail.com	DF-01
7	Child	7	C7	c7@gmail.com	P-104

update U-5-1

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	5	DF-01	Default	DF@gmail.com
2	Parent	2	P-102	P2	p2@gmail.com
3	Parent	3	P-103	P3	p3@gmail.com
4	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	DF-01
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	DF-01
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	DF-01
7	Child	7	C7	c7@gmail.com	P-104

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	5	DF-01	Default	DF@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com
4	Parent	2	U-501	P2	p2@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	DF-01
2	Child	2	C2	c2@gmail.com	DF-01
3	Child	3	C3	c3@gmail.com	DF-01
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	DF-01
6	Child	6	C6	c6@gmail.com	DF-01
7	Child	7	C7	c7@gmail.com	P-104

set default value

CASCADE

```sql
--Step: 6 CASCADE
--now Parent table and child table both table drop and both table same data
  with created
--only child table modification

DROP TABLE Child;

DROP TABLE Parent;
```

```sql
CREATE TABLE Parent(
    Parent_Id INT IDENTITY(1,1),
    Parent_Code NVARCHAR(5)
        CONSTRAINT PK_Tb_Parent_Col_Parent_Code PRIMARY KEY,
    Parent_Name NVARCHAR(15),
    Parent_Email NVARCHAR(20)
);
```

```sql
CREATE TABLE Child(
    Child_Id INT IDENTITY(1,1),
    Child_Name NVARCHAR(15),
    Child_Email NVARCHAR(20),
    Parent_Code NVARCHAR(5)
        CONSTRAINT FK_Tb_Child_Col_Parent_Code_PK_Tb_Parent_Col_Parent_Code
        FOREIGN KEY (Parent_Code) REFERENCES Parent(Parent_Code)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

```sql
SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	1	P-101	P1	p1@gmail.com
2	Parent	2	P-102	P2	p2@gmail.com
3	Parent	3	P-103	P3	p3@gmail.com
4	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	P-101
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	P-101
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	P-101
7	Child	7	C7	c7@gmail.com	P-104

```sql
DELETE FROM Parent WHERE Parent_Id = 1;

SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	P-102	P2	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	2	C2	c2@gmail.com	P-102
2	Child	4	C4	c4@gmail.com	P-103
3	Child	5	C5	c5@gmail.com	P-102
4	Child	7	C7	c7@gmail.com	P-104

delete

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	1	P-101	P1	p1@gmail.com
2	Parent	2	P-102	P2	p2@gmail.com
3	Parent	3	P-103	P3	p3@gmail.com
4	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	1	C1	c1@gmail.com	P-101
2	Child	2	C2	c2@gmail.com	P-102
3	Child	3	C3	c3@gmail.com	P-101
4	Child	4	C4	c4@gmail.com	P-103
5	Child	5	C5	c5@gmail.com	P-102
6	Child	6	C6	c6@gmail.com	P-101
7	Child	7	C7	c7@gmail.com	P-104

delte

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	P-102	P2	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	2	C2	c2@gmail.com	P-102
2	Child	4	C4	c4@gmail.com	P-103
3	Child	5	C5	c5@gmail.com	P-102
4	Child	7	C7	c7@gmail.com	P-104

```sql
UPDATE Parent
SET Parent_Code = 'H-501'
WHERE Parent_Id = 2;

SELECT 'Parent' AS 'Table_Name',* FROM Parent;
SELECT 'Child' AS 'Table_Name',* FROM Child;
```

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	H-501	P2	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	2	C2	c2@gmail.com	H-501
2	Child	4	C4	c4@gmail.com	P-103
3	Child	5	C5	c5@gmail.com	H-501
4	Child	7	C7	c7@gmail.com	P-104

update H-501

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	P-102	P2	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	2	C2	c2@gmail.com	P-102
2	Child	4	C4	c4@gmail.com	P-103
3	Child	5	C5	c5@gmail.com	P-102
4	Child	7	C7	c7@gmail.com	P-104

Results | Messages

	Table_Name	Parent_Id	Parent_Code	Parent_Name	Parent_Email
1	Parent	2	H-501	P2	p2@gmail.com
2	Parent	3	P-103	P3	p3@gmail.com
3	Parent	4	P-104	P4	p4@gmail.com

	Table_Name	Child_Id	Child_Name	Child_Email	Parent_Code
1	Child	2	C2	c2@gmail.com	H-501
2	Child	4	C4	c4@gmail.com	P-103
3	Child	5	C5	c5@gmail.com	H-501
4	Child	7	C7	c7@gmail.com	P-104

update new value